Excerpts

1 Analytic Equations

The famous formula of Leibniz involving π is

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots = \sum_{k \geq 0} \frac{(-1)^k}{2k+1}. \tag{1}$$

However, (1) converges far too slowly to be a practical method of approximating π.

Euler's integral formula for $n!$ is

$$n! = \int_0^\infty x^n e^{-x}\, dx.$$

The right side makes sense at $n = 1/2$ and can be evaluated explicitly:

$$\int_0^\infty \sqrt{x} e^{-x}\, dx = \frac{\sqrt{\pi}}{2}.$$

This suggests defining $(1/2)! = \dfrac{\sqrt{\pi}}{2}$.

For $a \in \mathbb{R}$ and a function $f\colon \mathbb{R} \to \mathbb{R}$, the *derivative* of f at a is

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \end{aligned}$$

and the formula for differentiation under the integral sign says that under suitable conditions,

$$\frac{d}{dt} \int_a^b f(x,t)\, dx = \int_a^b \frac{\partial f}{\partial t}(x,t)\, dx.$$

More formulas like this can be found in the book [Rud76].

2 Areas

If we are dealing with lattice polygons that are not simple, then we can use the following extension of Pick's Theorem.

Theorem 2.1. *For a lattice polygon P containing h holes with a boundary of $h+1$ simple closed curves, its area is given by*

$$A(P) = I(P) + \frac{B(P)}{2} + h - 1,$$

where $I(P)$ is the number of interior lattice points of P and $B(P)$ is the number of boundary lattice points of P.

We now review the definition of Euler characteristic for a convex polyhedron.

Definition 2.2. The *Euler characteristic* of a convex polyhedron P with V vertices, E edges, and F faces is $\chi(P) = V - E + F$.

Next, we recall Euler's formula.

Theorem 2.3 (Euler's Formula)**.** *For any convex polygon P with Euler characteristic $\chi(P)$, we have that*

$$\chi(P) = 2. \tag{2}$$

We will now generalize Theorem 2.1 by incorporating the Euler characteristic, which will allow us to find the area $A(P)$ of any lattice polygon P that is the union of a finite number of simple lattice polygons. This more general area formula is given by

$$A(P) = V - \frac{1}{2}E_B - \chi(P), \tag{3}$$

where V is the total number of vertices (lattice points) of P, where E_B is the number of boundary points on the edges of P, and where $\chi(P)$ is the Euler characteristic of P.

Note that, by Definition 2.2, the Euler characteristic of a simple lattice polygon (one with no holes) is $\chi(P) = 1$ and, by definition, $E_B = B(P)$. Furthermore, the Euler characteristic of a lattice polygon P containing h separate holes can be shown to be $\chi(P) = 1 - h$. We will now use Equation (2) to prove the generalized area formula given by Equation (3).

3 Matrix Equations

Operations on diagonal matrices are easy. For instance,

$$D = \begin{pmatrix} x & 0 & 0 \\ 0 & y & 0 \\ 0 & 0 & z \end{pmatrix} \Longrightarrow D^3 = \begin{pmatrix} x^3 & 0 & 0 \\ 0 & y^3 & 0 \\ 0 & 0 & z^3 \end{pmatrix}.$$

The transpose of a row vector $\mathbf{a} = (a_1, a_2, \ldots, a_n)$ is

$$\mathbf{a}^\top = \begin{pmatrix} a_1 \\ a_2 \\ \vdots \\ a_{n-1} \\ a_n \end{pmatrix}.$$

Remark 3.1. If you want to use LaTeX to edit a math page on Wikipedia, look at [Wik19].

References

[Rud76] Walter Rudin. *Principles of mathematical analysis*. McGraw-hill New York, 1976.

[Wik19] Wikipedia. https://en.wikipedia.org/wiki/Help:Displaying_a_formula, 2019. Accessed: 21-January-2020.